

WELLAND-PELHAM, ON, Canada

WMO: 717520

Lat: 42.9733N

Lon: 79.3250W

Elev: 591

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.2	4.6	-8.4	3.6	-0.6	-2.8	4.8	5.8	25.0	27.6	22.6	27.6	6.5	260	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	85.3	71.2	83.1	70.1	81.0	69.1	74.4	81.1	73.0	79.6	71.6	78.0	8.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	122.0	78.6	70.8	115.9	77.2	69.3	110.2	75.9	38.3	81.2	37.0	79.8	35.8	78.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	17.6	15.5	DB	-10.8	89.4	7.4	1.9	-16.1	90.7	-20.4	91.8	-24.6	92.9	-30.0	94.3
			WB	-11.6	76.8	7.0	2.3	-16.6	78.5	-20.7	79.8	-24.6	81.0	-29.7	82.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	24.9	24.5	33.1	44.8	57.5	65.8	70.5	68.9	62.7	51.5	39.7	31.0	(6)
	DBStd	18.44	11.36	11.06	10.26	8.30	8.05	5.77	5.08	5.05	7.00	8.16	8.50	9.00	(7)
	HDD50	3240	778	715	533	196	22	0	0	0	1	81	323	591	(8)
	HDD65	6685	1242	1134	990	605	260	58	11	20	125	426	759	1055	(9)
	CDD50	2521	1	0	8	41	253	475	635	585	381	127	14	1	(10)
	CDD65	493	0	0	0	0	27	84	181	140	55	6	0	0	(11)
	CDH74	3497	0	0	2	11	258	543	1305	939	416	23	0	0	(12)
	CDH80	646	0	0	0	0	41	96	267	159	81	2	0	0	(13)
(14) Wind	WSAvg	7.0	8.7	8.5	7.9	8.0	6.5	6.0	5.2	5.1	5.2	6.9	7.5	8.2	(14)
(15) Precipitation	PrecAvg	38.1	3.1	2.1	2.7	3.3	3.3	2.6	3.3	3.1	3.8	3.5	3.7	3.0	(15)
	PrecMax	45.9	5.3	4.4	5.0	5.6	6.1	6.7	7.3	5.6	7.5	7.8	5.5	6.1	(16)
	PrecMin	31.3	1.6	1.1	1.4	1.6	0.8	0.7	1.1	1.2	1.5	1.4	1.7	1.2	(17)
	PrecStd	4.7	1.1	0.7	1.0	1.1	1.5	1.5	1.7	1.2	1.5	1.6	0.9	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	55.1	68.1	74.8	84.1	85.7	88.2	87.6	86.0	77.0	65.5	57.1	(19)
		MCWB	51.5	48.7	58.9	60.4	68.6	68.7	72.1	70.7	72.1	66.3	56.5	52.7	(20)
	2%	DB	49.3	49.2	57.3	69.4	80.4	82.9	85.6	83.8	82.3	72.2	60.8	51.0	(21)
		MCWB	46.1	44.7	49.7	56.0	66.7	68.4	72.1	71.1	70.7	62.9	54.0	47.4	(22)
	5%	DB	44.4	43.2	52.6	63.8	76.6	79.9	83.0	81.5	78.9	68.6	57.3	47.3	(23)
		MCWB	41.3	39.6	46.9	53.0	63.8	67.0	70.4	70.0	68.0	61.9	50.7	44.3	(24)
	10%	DB	40.3	38.6	48.1	58.9	72.2	77.0	80.8	79.1	75.4	64.9	53.7	43.1	(25)
		MCWB	37.9	35.5	43.0	50.1	60.9	66.0	69.3	69.2	67.0	59.2	48.8	40.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.1	59.4	61.9	71.1	73.1	77.4	76.1	74.5	69.4	58.5	52.4	(27)
		MCDB	54.4	53.9	67.0	73.0	81.5	82.4	84.6	82.8	81.2	75.5	63.8	55.8	(28)
	2%	WB	46.1	44.8	51.8	57.8	68.2	71.1	75.0	74.1	72.6	66.0	55.2	47.6	(29)
		MCDB	48.2	48.3	55.5	66.6	78.5	79.1	81.5	80.4	79.0	69.5	60.2	49.9	(30)
	5%	WB	41.5	39.8	47.4	54.7	65.7	69.4	73.2	72.7	70.8	62.6	51.9	43.9	(31)
		MCDB	43.7	42.7	52.0	61.2	74.5	76.3	80.0	78.6	76.9	67.5	56.2	46.8	(32)
	10%	WB	37.6	35.5	43.2	51.2	62.9	67.8	71.7	71.1	68.5	59.5	49.1	39.9	(33)
		MCDB	40.0	38.3	47.7	57.2	69.9	74.3	78.1	76.9	74.1	64.2	53.5	42.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.4	15.4	18.3	19.8	17.4	17.5	17.6	18.3	15.7	14.2	11.3	(35)
		MCDBR	15.4	19.7	22.3	27.1	24.7	21.3	19.3	19.3	21.3	19.7	20.0	15.3	(36)
	5% WB	MCWBR	14.2	16.6	15.8	16.8	12.8	10.7	9.4	9.8	11.5	12.3	14.6	14.0	(37)
		MCWBR	14.8	19.0	20.3	23.3	22.2	16.6	15.8	15.7	18.2	16.8	18.2	15.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.298	0.333	0.335	0.377	0.405	0.435	0.436	0.422	0.382	0.362	0.314	0.300	(40)
		taud	2.410	2.303	2.397	2.322	2.292	2.241	2.259	2.285	2.375	2.409	2.509	2.462	(41)
	Ebn at Noon		266	272	288	282	276	267	265	265	268	259	259	255	(42)
		Edh at Noon	24	32	33	39	41	43	42	40	34	29	22	22	(43)
(44) All-Sky Solar Radiation	RadAvg		424	672	1042	1400	1716	1864	1874	1659	1318	799	495	335	(44)
	RadStd		39	62	119	149	179	138	142	89	121	76	69	42	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page